

LOVELY PROFESSIONAL UNIVERSITY

Summer PEP Test 01

ECMAScript 6 (ES6) - Question Paper

Course/Subject: JavaScript / ECMAScript 6	Test: Summer PEP Test 01
Level: Medium to Hard	Duration: _____
Maximum Marks: _____	Date: _____
Name: _____	Registration No.: _____

General Instructions

1. Attempt all questions.
2. This question paper contains 30 multiple choice questions and 2 coding questions.
3. Each MCQ has only one correct option.
4. Coding questions must be submitted through a public or private GitHub repository as instructed.
5. Write clean, readable, and properly commented JavaScript code.
6. Use ES6 features wherever applicable, especially let, const, arrow functions, template literals, map(), filter(), and reduce().
7. Do not write answers on this question paper unless instructed by the invigilator.

Section A: Multiple Choice Questions

Choose the most appropriate option for each question.

1. Which statement best describes ECMAScript 6?

- A. It is a separate programming language from JavaScript.
- B. It is the 2015 standard version of JavaScript with major language improvements.
- C. It is a library used only for frontend development.
- D. It is a database query language used with Node.js.

2. What will be the output of the following code?

```
console.log(a);  
let a = 10;
```

- A. undefined
- B. 10
- C. ReferenceError
- D. TypeError

3. What will be the output of the following code?

```
console.log(x);  
var x = 20;
```

- A. undefined
- B. 20
- C. ReferenceError
- D. TypeError

4. Which option correctly explains the Temporal Dead Zone in JavaScript?

- A. The time delay caused by asynchronous functions.
- B. The area where function declarations are stored before execution.
- C. The period between entering a block scope and declaring a let/const variable.
- D. The memory area where unused variables are deleted.

5. What will be printed by the following code?

```
let score = 5;
{
  let score = 10;
}
console.log(score);
```

- A. 5
- B. 10
- C. undefined
- D. ReferenceError

6. What will be printed by the following code?

```
var total = 5;
{
  var total = 10;
}
console.log(total);
```

- A. 5
- B. 10
- C. undefined
- D. ReferenceError

7. Which statement about const is correct?

- A. A const variable can be declared without initialization.
- B. A const object cannot have its properties changed.
- C. A const variable cannot be reassigned after initialization.
- D. A const variable has function scope like var.

8. What will be the output of the following code?

```
const user = { name: 'Amit' };
user.name = 'Ravi';
console.log(user.name);
```

- A. Amit
- B. Ravi
- C. TypeError
- D. ReferenceError

9. Which statement is true about arrow functions?

- A. They always require the function keyword.
- B. They have their own this binding.
- C. They are suitable for shorter function expressions.
- D. They can always be used as constructors with new.

10. What will be the output?

```
const greet = name => `Hello ${name}`;
console.log(greet('Riya'));
```

- A. Hello undefined
- B. Hello Riya
- C. TypeError
- D. ReferenceError

11. Which arrow function syntax is correct for returning the square of a number?

- A. const square = x => x * x;
- B. const square => x = x * x;
- C. const square = function => x * x;
- D. const square(x) => x * x;

12. What is the major difference between a regular function and an arrow function regarding this?

- A. Arrow functions create a new this every time they run.
- B. Arrow functions inherit this from the surrounding lexical scope.
- C. Regular functions cannot access this.
- D. Both always behave exactly the same.

13. What will be the output after one second?

```
for (let i = 0; i < 3; i++) {  
  setTimeout(() => console.log(i), 1000);  
}
```

- A. 0 1 2
- B. 3 3 3
- C. undefined undefined undefined
- D. ReferenceError

14. What will be the output after one second?

```
for (var i = 0; i < 3; i++) {  
  setTimeout(() => console.log(i), 1000);  
}
```

- A. 0 1 2
- B. 3 3 3
- C. 1 2 3
- D. ReferenceError

15. Which feature is used in the following code?

```
const [first, second] = [100, 200];
```

- A. Spread operator
- B. Array destructuring
- C. Rest parameter
- D. Template literal

16. What will be the output?

```
const [a, , c] = [10, 20, 30];  
console.log(c);
```

- A. 10
- B. 20
- C. undefined
- D. ReferenceError

17. Which feature is used in the following code?

```
const employee = { id: 101, name: 'Neha' };  
const { name } = employee;
```

- A. Array destructuring
- B. Object destructuring
- C. Spread operator
- D. Hoisting

18. What will be stored in copied?

```
const arr = [1, 2, 3];  
const copied = [...arr];
```

- A. A reference to the same array only
- B. A shallow copy of arr
- C. A deep copy of all nested objects
- D. An empty array

19. What is the purpose of the rest parameter in ES6?

- A. To spread array values into individual values.
- B. To collect remaining arguments into an array.
- C. To stop function execution.
- D. To copy an object deeply.

20. What will be the output?

```
function count(...values) {  
  return values.length;  
}  
console.log(count(10, 20, 30));
```

- A. 1
- B. 3
- C. 6
- D. [1, 2, 3]

21. Which method is best suited when you want to create a new array by transforming every element of an existing array?

- A. filter()
- B. map()
- C. reduce()
- D. find()

22. What will be the output?

```
const nums = [1, 2, 3];  
const result = nums.map(n => n * 2);  
console.log(result);
```

- A. [2, 4, 6]
- B. [1, 2, 3]
- C. 6
- D. [1, 4, 9]

23. Which method should be used to return only elements that satisfy a condition?

- A. map()
- B. filter()
- C. reduce()
- D. push()

24. What will be the output?

```
const marks = [5, 10, 15, 20];  
const selected = marks.filter(m => m > 10);  
console.log(selected);
```

- A. [5, 10]
- B. [15, 20]
- C. [10, 15, 20]
- D. 20

25. Which method is most appropriate to calculate a single final value from an array?

- A. map()
- B. filter()
- C. reduce()
- D. slice()

26. What will be the output?

```
const nums = [1, 2, 3, 4];  
const total = nums.reduce((acc, curr) => acc + curr, 0);  
console.log(total);
```

- A. 10
- B. 24
- C. [1, 2, 3, 4]
- D. undefined

27. In `reduce((acc, curr) => acc + curr, 0)`, what does 0 represent?

- A. The current array index
- B. The initial value of the accumulator
- C. The length of the array
- D. The final return value

28. What will happen in the following code?

```
const x;  
x = 10;  
console.log(x);
```

- A. It will print 10.
- B. It will print undefined.
- C. It will throw an error because const must be initialized.
- D. It will convert x into var automatically.

29. Which of the following is a valid use of template literals?

- A. 'Hello \${name}'
- B. `Hello \${name}`
- C. "Hello \$name"
- D. template Hello(name)

30. Which option shows a correct ES6 approach to combine two arrays without modifying the original arrays?

- A. `arr1.push(arr2)`
- B. `const result = arr1 + arr2`
- C. `const result = [...arr1, ...arr2]`
- D. `const result = arr1.join(arr2)`

Section B: Coding Questions

Both coding questions are compulsory. Students must submit the GitHub repository link as per the instructions given below.

Coding Question 1: Student Performance Analyzer

Problem Statement:

A faculty member wants to analyze student performance after a JavaScript mid-term evaluation. You are required to build a console-based ES6 program that processes student records using array methods.

Given Data:

```
const students = [
  { id: 1, name: 'Aman', marks: 85, attendance: 92 },
  { id: 2, name: 'Riya', marks: 38, attendance: 76 },
  { id: 3, name: 'Karan', marks: 67, attendance: 81 },
  { id: 4, name: 'Simran', marks: 91, attendance: 88 },
  { id: 5, name: 'Raj', marks: 45, attendance: 60 },
  { id: 6, name: 'Neha', marks: 29, attendance: 95 }
];
```

Tasks to Perform:

1. Use filter() to create a list of students who have passed. Passing marks are 40 or above.
2. Use map() to generate a new array containing each student's name, marks, attendance, and grade.
3. Grade rules: marks $\geq$ 80: A, marks $\geq$ 60: B, marks $\geq$ 40: C, otherwise: F.
4. Use reduce() to calculate the class average marks.
5. Use reduce() to find the topper of the class.
6. Display the final results using template literals in a neat format.
7. Use let, const, arrow functions, destructuring, and meaningful variable names.

Expected Output Requirements:

- List of passed students.
- Student report with calculated grades.
- Class average marks.
- Topper's name and marks.

Coding Question 2: Online Cart Billing System

Problem Statement:

An online store wants a billing system that calculates product-wise totals, removes unavailable items, applies discount rules, and produces a final bill. Build this as a console-based ES6 JavaScript program.

Given Data:

```
const cart = [
  { productId: 101, name: 'Laptop', price: 55000, quantity: 1, category: 'Electronics' },
  { productId: 102, name: 'Mouse', price: 700, quantity: 2, category: 'Accessories' },
  { productId: 103, name: 'Keyboard', price: 1500, quantity: 1, category: 'Accessories' },
  { productId: 104, name: 'Monitor', price: 12000, quantity: 0, category: 'Electronics' },
  { productId: 105, name: 'USB Cable', price: 300, quantity: 3, category: 'Accessories' },
  { productId: 106, name: 'Office Chair', price: 6500, quantity: 1, category: 'Furniture' }
];
```

Tasks to Perform:

1. Use filter() to remove products whose quantity is 0.
2. Use map() to create a new array containing product name, quantity, price, and itemTotal.
3. Use reduce() to calculate the subtotal amount of the cart.
4. Apply a 10% discount if the subtotal is greater than 50000; otherwise apply no discount.
5. Add 18% GST after discount and calculate the final payable amount.
6. Use reduce() to calculate category-wise total amount.
7. Display a complete bill summary using template literals.

8. Use ES6 features such as const, let, arrow functions, spread/rest where useful, and clean formatting.

Expected Output Requirements:

- Available products only.
- Product-wise bill details.
- Subtotal, discount amount, GST amount, and final payable amount.
- Category-wise total amount.

GitHub Submission Guidelines

1. Create a GitHub repository for each coding question or one combined repository containing separate folders for both questions.
2. Each folder must contain index.js and README.md.
3. README.md must include project title, problem statement, ES6 concepts used, steps to run, sample output, output screenshot, and GitHub repository link.
4. Run the program using: node index.js
5. Upload the output screenshot in the repository and mention the file path in README.md.
6. Submit only the GitHub repository link through the assigned submission portal.

- End of Question Paper -